CALEB UNIVERSITY, LAGOS · B.Sc. COMPUTER SCIENCE · FINAL YEAR PROJECT

Design and Development of a Nigerian Public Economic Data Aggregation and Analytics Platform with Open API Access

A 2020–2026 Case Study · "NPEDATA"

By: Taoheed Abdulmanan Olaosebikan (22/10267)
Supervisor: Miss Ilori Deborah
Head of Department: Dr. Ayorinde Oduroye
Department: Computer Science, College of Science and Information Science (COSIS)
Date: July 2026

INTRODUCTION

Background of the Study

Reliable data exists, but its published form makes it hard to use.

- **Data drives decisions** — government, researchers, journalists and the public all rely on economic figures.
- **Nigeria's sources are credible** — the figures come from the CBN, the NBS and the World Bank.
- **But the data is scattered** — it is spread across many websites as PDFs and spreadsheets.
- **So it is hard to use** — any analysis must begin by collecting and cleaning the data by hand.

THE PROBLEM

Statement of the Problem

Four barriers make Nigeria's public data hard to use in practice.

- **Not machine-readable** — the data is locked inside PDF documents.
- **No common structure** — units and time periods differ from one source to the next.
- **No open API** — there is no free way to query the data with a program.
- **Repeated effort** — every user must redo the same collection and cleaning.

Aim and Objectives

Aim: To build a platform that gathers Nigeria's public economic data into one standard store, available through a dashboard and a free, open API.

1. Collect economic indicators from the CBN, NBS and World Bank into one repository.
2. Standardise them into one relational database model.
3. Provide analytics: change, trend and correlation.
4. Build a free REST API with HATEOAS and no login.
5. Build a clear, interactive dashboard.
6. Test the system for correctness, usability and accessibility.

SCOPE

Scope of the Project

A clear boundary: what the project covers, and what it does not.

- **It is a prototype** — a working system with 122 indicators and about 12,100 records.
- **It is not a national system** — not official infrastructure, not a bank, not real-time.
- **It uses no AI** — the analytics are kept simple and transparent on purpose.
- **Data is collected manually** — the figures reflect the latest snapshot, not a live feed.

Key Concepts

The project applies established, well-known principles rather than inventing new ones.

- **Open data** — information is only useful when it is machine-readable.
- **Tidy data** — one record per row, with metadata kept separate (Wickham, 2014).
- **REST and HATEOAS** — an API that describes itself to its users (Fielding, 2000).
- **Honest charts** — avoid misleading tricks such as dual axes (Tufte, 1983).

EXISTING SYSTEMS

Related Systems and the Gap

Good systems exist, but none meet all three needs at once.

- **FRED and the World Bank** — excellent access, but little detailed Nigerian data.
- **Trading Economics, Statista** — mostly paywalled, and resell the same figures.
- **The CBN and NBS** — reliable sources, but they offer no API.
- **The gap** — no system offers Nigerian data, open access and honesty together.

MOTIVATION

Why This Has Not Been Built

The obstacle is mainly institutional, not technical (all points are cited).

- **A framework already exists** — a government interoperability framework since 2018.
- **Data is treated as an asset** — agencies are often reluctant to share it (Eleanya, 2026).
- **Rules are not enforced** — 153 of 250 agencies fell below the FOI benchmark (Ogunyale & Osho, 2023).
- **This project shows it is doable** — the technical barrier is low.

METHODOLOGY

Research Methodology

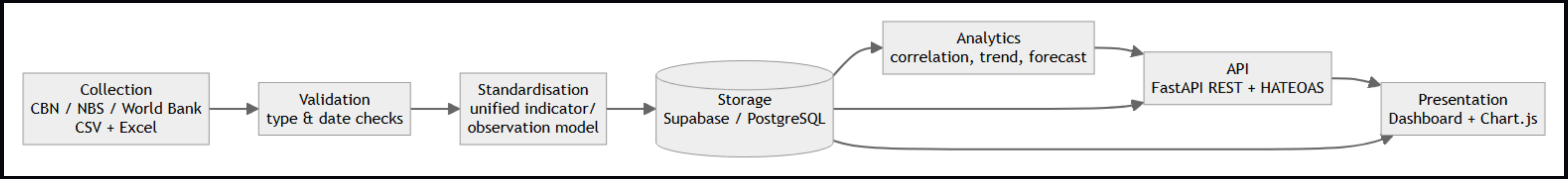
An iterative approach suited a solo project with requirements that emerged over time.

- **Built in stages** — data model first, then API, then dashboard, then analytics.
- **Checked every stage** — displayed figures were verified against the database each cycle.
- **Not Waterfall or Scrum** — requirements were not fully known, and this was a one-person project.

SYSTEM DESIGN

System Architecture

A seven-stage pipeline transforms the data from source to screen.



Collect → Validate → Standardise → Store → Analyse → API → Present.

SYSTEM DESIGN

Database Design

One tidy table can hold series of every time frequency.

- **Three linked tables** — sources, indicators and observations.
- **One observations table** — stores every series as (indicator, date, value).
- **Built-in integrity** — keys and a uniqueness rule prevent duplicate records.
- **Units are stored** — each series keeps its unit, avoiding thousands-vs-millions errors.

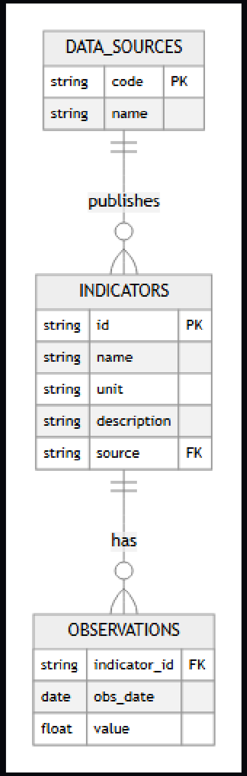


Figure 3.4 — Entity-relationship diagram.

IMPLEMENTATION

The Open API

A free, self-describing API — the machine access the original sources lack.

- **Built with FastAPI** — 17 endpoints, and no login is required.
- **Self-describing** — every response includes links to related data (HATEOAS Level 3).
- **Fully navigable** — the whole API can be explored from one starting point.
- **Ready for machines** — it also serves AI agents through an MCP server.

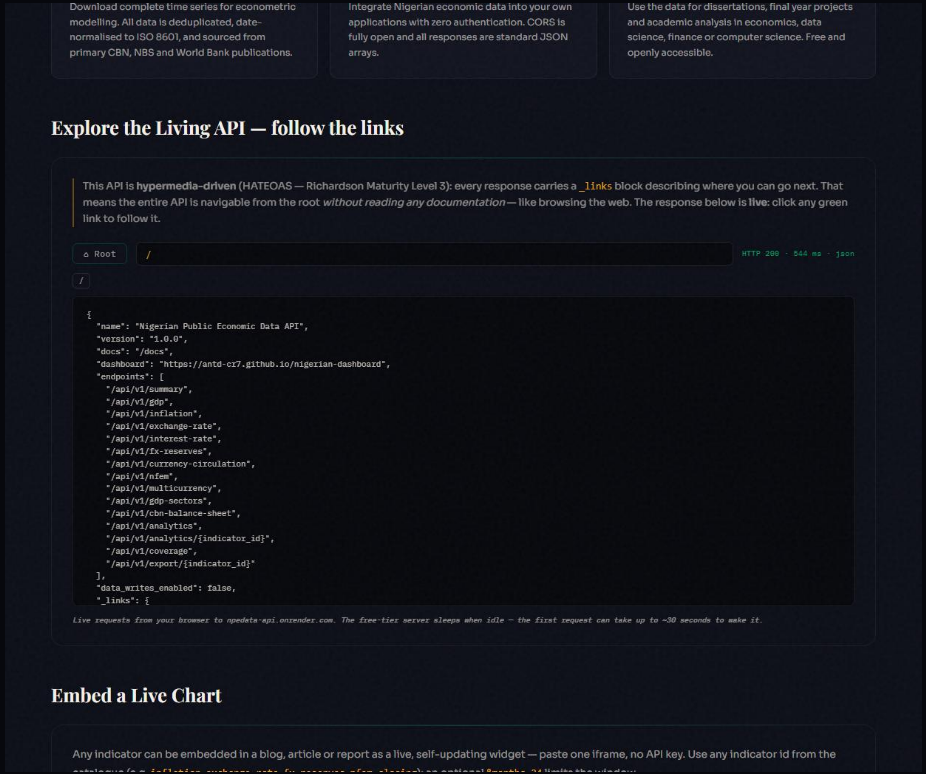


Figure 4.11 — HATEOAS Explorer browsing the live API.

IMPLEMENTATION

The Dashboard

One clear interface that serves both the public and researchers.

- **Each page tells a story** — what happened, why it matters, and how to read it.
- **A Reader / Analyst switch** — the same page serves the public and researchers.
- **Honest charts** — no misleading dual axes, and units are always shown.
- **Fully accessible** — it scores 100 out of 100 for accessibility.

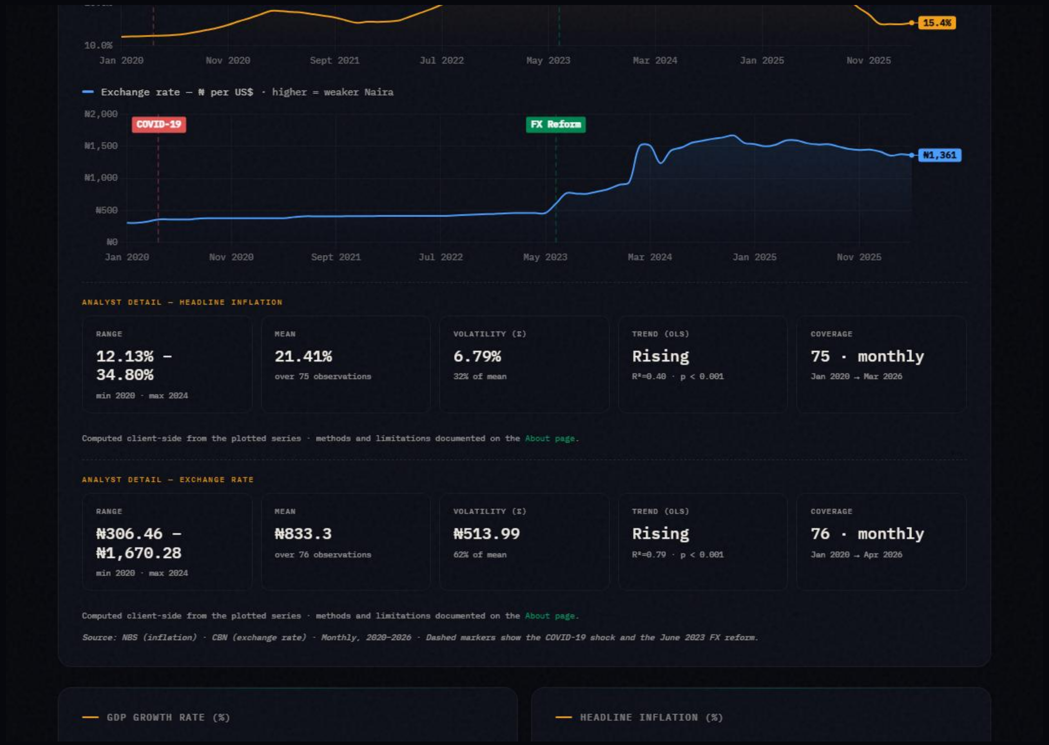


Figure 4.4 — The Reader/Analyst dial revealing statistical detail.

ANALYTICS

The Analytics Engine

The platform measures relationships in the data, not just displays numbers.

- **A full profile for any indicator** — latest value, change, range, volatility and trend.
- **Correlation with significance** — shown with an R^2 value and a p-value.
- **A correlation matrix** — compares 12 key indicators against each other at once.
- **Transparent methods** — all classical statistics, no hidden AI model.

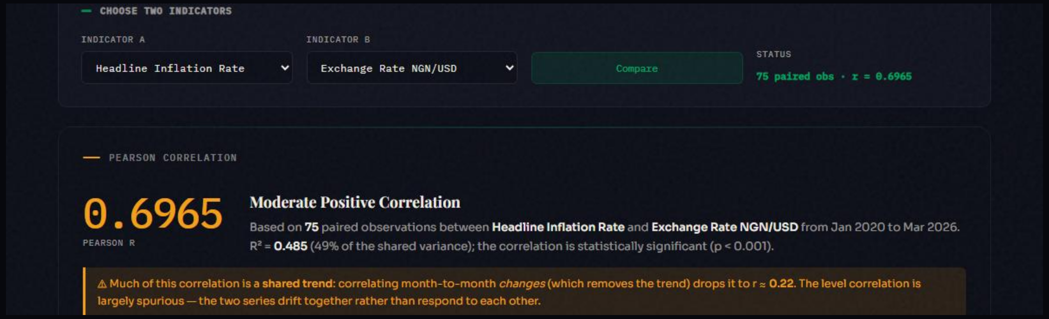


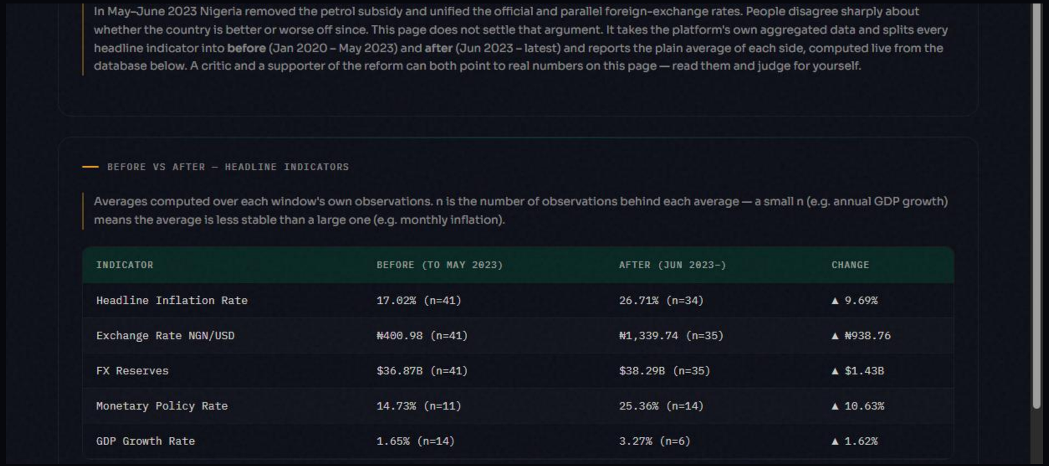
Figure 4.9 — Correlation reported with R^2 and a significance p-value.

FEATURE

Reform Impact

The data applied to a real national debate, without taking a side.

- **Before and after June 2023** — it compares every indicator across the reforms.
- **Computed from the data** — the figures are calculated live from the stored records.
- **It takes no side** — critics and supporters see the very same numbers.



In May–June 2023 Nigeria removed the petrol subsidy and unified the official and parallel foreign-exchange rates. People disagree sharply about whether the country is better or worse off since. This page does not settle that argument. It takes the platform's own aggregated data and splits every headline indicator into **before** (Jan 2020 – May 2023) and **after** (Jun 2023 – latest) and reports the plain average of each side, computed live from the database below. A critic and a supporter of the reform can both point to real numbers on this page — read them and judge for yourself.

BEFORE VS AFTER — HEADLINE INDICATORS

Averages computed over each window's own observations. n is the number of observations behind each average — a small n (e.g. annual GDP growth) means the average is less stable than a large one (e.g. monthly inflation).

INDICATOR	BEFORE (TO MAY 2023)	AFTER (JUN 2023–)	CHANGE
Headline Inflation Rate	17.02% (n=41)	26.71% (n=34)	▲ 9.69%
Exchange Rate NGN/USD	₦400.98 (n=41)	₦1,339.74 (n=35)	▲ ₦938.76
FX Reserves	$36.87B (n=41)	$38.29B (n=35)	▲ $1.43B
Monetary Policy Rate	14.73% (n=11)	25.36% (n=14)	▲ 10.63%
GDP Growth Rate	1.65% (n=14)	3.27% (n=6)	▲ 1.62%

Figure 4.15 — Reform Impact: before and after June 2023, neutral readings.

INTEGRITY

Honesty and Validation

The platform flags results that could mislead the reader.

- **Warns about false links** — it flags a correlation caused only by a shared trend.
- **Separates two ideas** — statistical significance is kept apart from strength.
- **Public data-check tool** — anyone can validate a file, and it can never change the data.

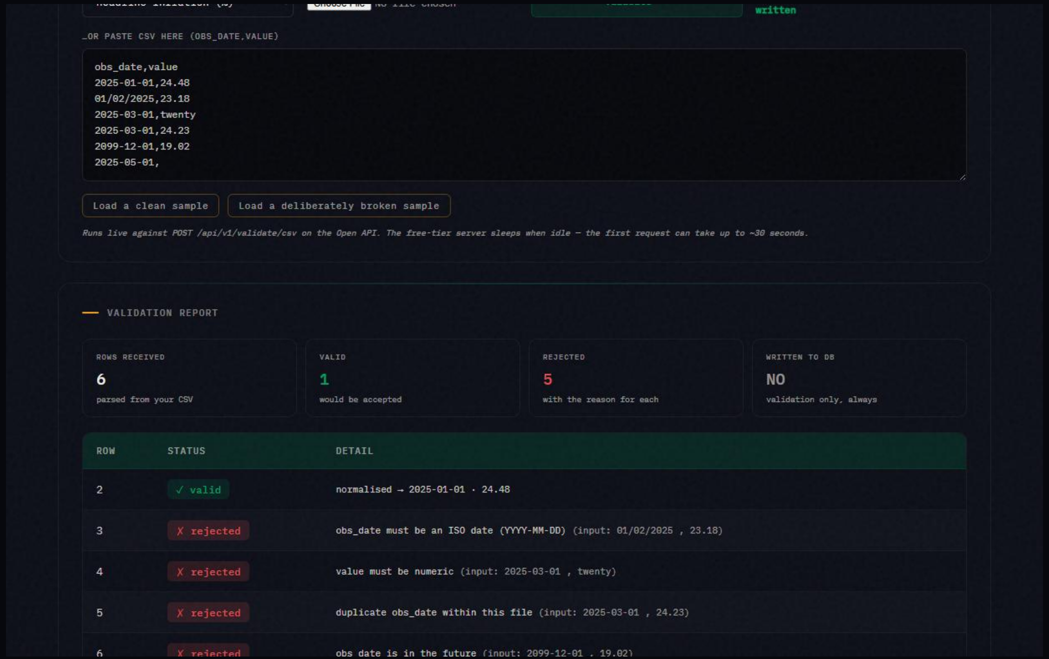


Figure 4.12 — Pipeline Playground: per-row verdicts, nothing written.

Testing and Validation

Correctness was tested and shown, not simply assumed.

- **24 automated tests** — they cover the API and all pass.
- **Statistics checked** — results were compared against known cases.
- **Data audit** — a review of the figures found and fixed real errors.
- **Handles bad input** — the system stays stable with invalid or extreme data.

RESULTS

Results

Clear, verifiable outputs that meet each objective.

- **122 indicators, 12,100 records** — all held in one store you can query.
- **17 live API endpoints** — documented and at HATEOAS Level 3.
- **100 out of 100** — a perfect accessibility score.
- **Verified figures** — every displayed number checked against the database.

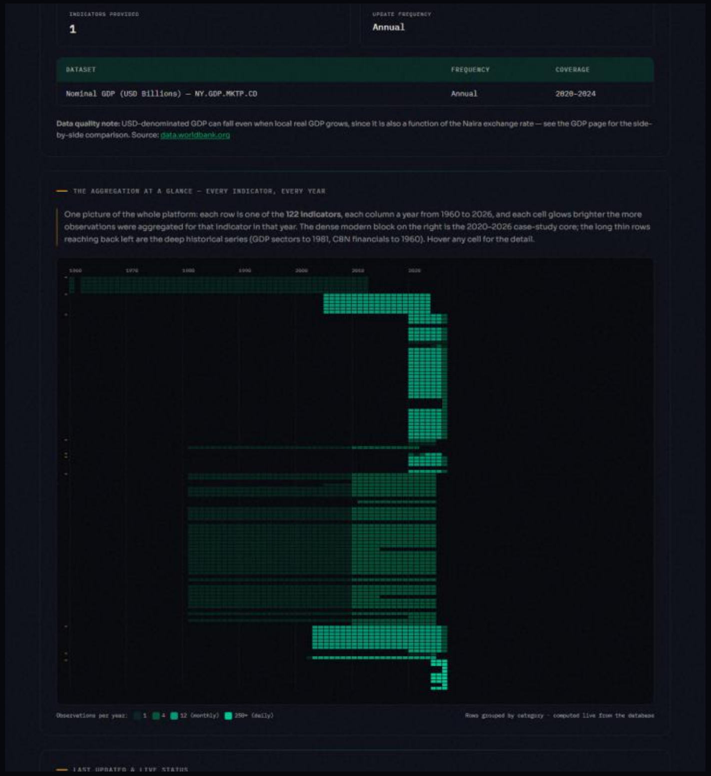


Figure 4.10 — The aggregation at a glance: 122 indicators × 1960–2026.

CONCLUSION

Contribution to Knowledge

A reusable result that can be rebuilt from the source code alone.

- **A working reference platform** — for unified Nigerian public economic data.
- **A genuinely open API** — with no freely available equivalent found.
- **Built with free tools** — reproducible entirely from the repository.
- **Evidence for the argument** — that unifying this data was always achievable.

FUTURE WORK

Limitations and Future Work

Clear next steps that build on the current prototype.

- **Automate collection** — connect directly to the source websites.
- **Expand coverage** — add more indicators, including state-level data.
- **Add richer analytics** — such as seasonal adjustment and forecasting.
- **Add API keys** — with rate limits for heavier users.

Conclusion

Nigeria's scattered, non-machine-readable public economic data need not stay that way: it can be consolidated into a correct, programmatically usable resource using only free and open-source tools.

Dashboard: antd-cr7.github.io/nigerian-dashboard
Open API: npedata-api.onrender.com (docs at /docs)
Source: github.com/ANTD-CR7/nigerian-dashboard

Thank you. Questions & live demonstration.